

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2016-17

Botany**Paper: BOB-501 — Comparative Studies of Cryptogams**

Time: Three Hours

Full Marks: 70

Note: Attempt five questions in all, selecting one question each from Section-A, B, C, and D. Section-E is compulsory. The figures in the right-hand margin indicate marks.

SECTION – A

Question 1. Give an illustrated account of thallus organization in Chlorophyceae. [15]

Question 2. Write explanatory notes on any two of the following: [7.5 × 2 = 15]

- (a) Triphasic life history in a red alga
- (b) Photosynthetic pigments in major classes of algae
- (c) Oogamous reproduction in any brown alga

SECTION – B

Question 3. With the help of suitable diagrams, give a critical account of types of fungal spores and modes of their liberation. [15]

Question 4. Write notes on any two of the following: [7.5 × 2 = 15]

- (a) Classification of Ascomycotina
- (b) Major types of life cycles in fungi
- (c) Importance of fungi in industry and agriculture

SECTION – C

Question 5. Give an illustrated account of evolution of sporophyte in Bryophyta by progressive sterilization of sporogenous tissues. [15]

Question 6. Write notes on any two of the following: [7.5 × 2 = 15]

- (a) Vegetative propagation in Bryophyta
- (b) Structure and anatomy of gametophytes of *Pellia* and *Notothylas*
- (c) Life history of *Sphagnum*

SECTION – D

Question 7. Give a critical account of structure of *Marsilea* sporocarp. [15]

Question 8. Write notes on any two of the following: [7.5 × 2 = 15]

- (a) Major types of steles in Pteridophyta
- (b) Apogamy and apospory in ferns
- (c) Salient features of *Psilotum*

SECTION – E

Question 9. Answer the following in not more than 50 words each: [1 × 10 = 10]

- (a) Name two important food reserve products found in Cyanophyceae.
- (b) Give one example each of algae showing isomorphic and heteromorphic alternation of generations.
- (c) Give examples of uniaxial and multiaxial thallus in algae.
- (d) What is diplanetism?
- (e) Name different types of spores produced by *Puccinia*.
- (f) Name a bryophyte which has elaterophores in the sporogonium.
- (g) What are retort cells?
- (h) Write down the name of a ligulate and heterosporous pteridophyte.
- (i) What is the function of peristome teeth?
- (j) What is false indusium?

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